

SOFTWARE REQUIREMENTS SPECIFICATION

HOTEL BOOKING SYSTEM

Complete SRS Document

Functional, non-functional, interface, database and system requirements for the
Hotel Booking System.

1. Introduction

1.1 Purpose

The purpose of the **Hotel Booking System** is to provide an online platform for customers to search hotels, check room availability, book rooms, make payments, and manage their bookings. The system also helps hotel administrators manage rooms, customers, pricing, bookings, and billing efficiently.

1.2 Scope

- Customer registration and login
- Hotel and room search
- Checking room availability
- Online room booking
- Seasonal/dynamic pricing management
- Customer billing and online payment
- Booking cancellation and booking history
- Admin management of rooms, prices and customers
- Generation of booking, payment and revenue reports

1.3 Definitions and Abbreviations

Term	Description
SRS	Software Requirements Specification
Admin	Person who manages the hotel system
Customer	Person who books a hotel room
UI	User Interface
DB	Database
API	Application Programming Interface
Booking ID	Unique identification number for a booking

2. Overall Description

2.1 Product Perspective

The system follows a **Frontend** → **Backend/API** → **Database** architecture. The frontend allows customers and administrators to interact with the application, the backend processes business logic, and the database stores customer, room, booking, payment and pricing information.

2.2 Product Functions

- User registration and login
- Hotel/room search
- Room availability checking
- Room booking
- Price calculation
- Seasonal pricing
- Payment processing
- Invoice generation

- Booking cancellation
- Booking history
- Room management
- Customer management
- Report generation

2.3 User Classes

User	Main Responsibilities
Customer	Register/login, search rooms, check availability, book, pay, view/cancel bookings and invoice
Administrator	Manage rooms, availability, seasonal prices, bookings, customers, payments and reports

2.4 Operating Environment

Frontend: HTML, CSS, JavaScript / React

Backend: Node.js and Express.js

Database: MySQL / MongoDB

Server: Web server or cloud server

Browsers: Chrome, Firefox, Edge and Safari

Operating systems: Windows, Linux, Android and iOS

3. Functional Requirements

FR1 – User Registration

The system shall allow customers to create accounts using name, email, phone number and password.

FR2 – User Login

The system shall allow registered customers and administrators to log in using valid credentials.

FR3 – Hotel/Room Search

Customers shall search using location, check-in date, check-out date, guests and room type.

FR4 – Room Availability

The system shall display available rooms and prevent double booking.

FR5 – Room Details

The system shall display room number, type, capacity, facilities, price, images and availability.

FR6 – Room Booking

Customers shall select an available room and create a booking with a unique Booking ID.

FR7 – Seasonal Pricing

Administrators shall define normal, peak, festival and holiday prices based on booking dates.

FR8 – Billing

The system shall calculate room charges, taxes/other charges, discounts and final amount.

FR9 – Payment

Customers shall make payments through supported methods and the system shall update payment status.

FR10 – Booking Confirmation

After successful booking/payment, the system shall provide Booking ID, customer, room, dates and payment status.

FR11 – Cancellation

Customers shall cancel eligible bookings according to the hotel's cancellation policy.

FR12 – Booking History

Customers shall view current, previous and cancelled bookings.

FR13 – Admin Room Management

Administrators shall add, update, remove rooms and change availability and prices.

FR14 – Customer Management

Administrators shall view and manage registered customer information.

FR15 – Reports

Administrators shall generate booking, availability, cancellation, revenue, customer and payment reports.

4. Non-Functional Requirements

4.1 Performance

The system should respond quickly and support multiple users simultaneously.

4.2 Security

Passwords and customer/payment information must be protected; admin functions require authorization.

4.3 Reliability

The system should operate reliably, prevent loss of booking information and maintain backups.

4.4 Usability

The interface should be simple, user-friendly and provide clear error messages.

4.5 Availability

The system should be available for searching and booking subject to server and maintenance availability.

4.6 Maintainability

The software should be modular so developers can modify or add features easily.

4.7 Scalability

The system should support increasing customers, hotels, rooms, bookings and transactions.

5. External Interface Requirements

5.1 User Interface

Home, Login, Registration, Search, Room Details, Booking, Payment, Booking Confirmation, Customer Dashboard and Admin Dashboard.

5.2 Hardware Interface

The system can be accessed using desktop computers, laptops, smartphones and tablets.

5.3 Software Interface

The system may interact with a database server, payment gateway, email/SMS service and web/application server.

5.4 Communication Interface

Secure communication such as HTTPS shall be used between client and server.

6. System Requirements

6.1 Hardware Requirements

Server: Dual Core processor or higher, 4 GB RAM or higher, 50 GB storage or higher.

Client: Smartphone, tablet, laptop or desktop with internet connection.

6.2 Software Requirements

Operating System: Windows/Linux/Android/iOS

Frontend: HTML, CSS, JavaScript/React

Backend: Node.js, Express.js

Database: MySQL/MongoDB

Browser: Chrome/Firefox/Edge/Safari

7. Database Requirements

Entity	Important Fields
Customer	Customer_ID, Name, Email, Phone, Password
Room	Room_ID, Room_Number, Room_Type, Capacity, Price, Availability_Status
Booking	Booking_ID, Customer_ID, Room_ID, Check_In, Check_Out, Number_of_Guests, Booking_Status, Total_Amount
Payment	Payment_ID, Booking_ID, Payment_Date, Amount, Payment_Method, Payment_Status
Seasonal Pricing	Pricing_ID, Season_Name, Start_Date, End_Date, Room_Type, Price

8. System Workflow

Customer: Register/Login → Search Hotel/Room → Check Availability → Select Room → Calculate Price → Enter Booking Details → Make Payment → Booking Confirmation → Generate Invoice.

Admin: Admin Login → Dashboard → Manage Rooms → Manage Availability → Manage Seasonal Pricing → Manage Bookings → Manage Customers → View Payments & Reports.

9. Use Case Summary

Actor	Use Cases
Customer	Register, Login, Search Rooms, Check Availability, Book Room, Make Payment, View Booking, Cancel Booking, View Booking Details
Admin	Login, Manage Rooms, Manage Pricing, Manage Bookings, Manage Customers, Generate Reports

10. Constraints

- Internet connection is required for online booking.
- Payment depends on payment gateway availability.
- Room availability must be updated accurately.
- Only authorized administrators can modify hotel information.
- The system must follow applicable data-protection and payment-security requirements.

11. Assumptions and Dependencies

11.1 Assumptions

- Customers have valid contact information.
- Hotel room information is maintained by the administrator.
- Customers provide correct check-in and check-out dates.
- The payment service is available when payment is attempted.

11.2 Dependencies

- Internet connection
- Database server
- Web/application server
- Payment gateway
- Email/SMS service, if implemented

12. Expected Outcome

The Hotel Booking System provides an efficient way to manage hotel room reservations, availability, seasonal pricing and customer billing. It reduces manual booking work, helps avoid double booking, and provides customers with a convenient online booking experience.

13. Conclusion

The proposed Hotel Booking System integrates room management, availability checking, booking, seasonal pricing, payment and billing into a single system. This SRS defines the functional and non-functional requirements needed to develop the system in a structured and reliable manner.